

Notes: Borella, De Nardi & Yang (RES, 2023)

Are Marriage-Related Taxes and Social Security Benefits Holding Back Female Labour Supply?

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1 Big picture and incremental contribution

- **Question.** How much do U.S. marriage-based income taxation and Social Security spousal/survivor benefits depress female labour supply and alter savings and welfare?
- **Mechanism.** Joint taxation raises the marginal tax rate of secondary earners; spousal and survivor benefits weaken the link between own labour supply and retirement income.
- **Approach.** The paper estimates a dynamic life-cycle model for couples and singles with labour supply, savings, marriage/divorce, wage shocks, human capital, retirement, health, medical expenses, taxes, Social Security, EITC, Medicaid, and SSI.
- **Main result.** Removing marriage-related provisions raises married women's participation, raises single women's participation when they expect marriage, lowers older married men's participation, and raises retirement wealth.
- **Welfare result.** Balanced-budget reforms generate large gains for most couples and single men and for many single women.

1.1 Incremental contribution of the paper

- It studies all marriage-related tax and benefit provisions in one framework, rather than studying joint taxation or Social Security spousal benefits separately.
- It estimates the model for two cohorts, 1945 and 1955, allowing the paper to ask whether the same policies still matter after female participation has risen.
- It is a dynamic model of both couples and singles in which both men and women choose participation and hours and households save.
- It includes human-capital accumulation, so a labour-supply wedge today lowers future wages.
- It explicitly models Social Security claiming, spousal benefits, survivor benefits, health, survival, medical expenses, Medicaid, and SSI.
- It evaluates welfare, not only hours or participation.

Interpretation. The paper's contribution is a dynamic, quantitatively disciplined accounting of how marriage-based policy rules affect work, experience, retirement insurance, savings, and welfare jointly.

2 Data and measurement

2.1 Cohorts and data sources

The 1945 cohort is born in 1941–1945 and is observed through working and retirement ages. The 1955 cohort is born in 1951–1955 and has higher married-women participation. The PSID supplies working-age participation, hours, wages, wealth, marriage, divorce, matching, fertility, and tax functions. The HRS supplies survival, health transitions, and out-of-pocket medical expenses.

Interpretation. The two cohorts provide a test of whether marriage provisions still bind when women's labour-market attachment is stronger.

2.2 Target moments and estimated inputs

Targeted moments include participation, hours among workers, and wealth for married women, married men, single women, and single men. First-step inputs include wage processes, marriage/divorce, matching, children, survival, health transitions, medical expenses, and policy rules.

Interpretation. The model is disciplined by labour and savings moments across gender-marital groups, which is necessary because policy reforms affect both work and retirement wealth.

2.3 Policy measurement

The policy environment includes progressive income taxation, joint filing, payroll taxes, EITC, own Social Security benefits, spousal benefits, survivor benefits, Medicaid, and SSI. Counterfactuals remove spousal/survivor benefits, joint taxation, or both. Balanced-budget experiments rebate government surplus by lowering the proportional component of the income tax.

Interpretation. The policy experiments isolate marriage-based wedges while keeping the government budget transparent.

3 Model and empirical specifications

3.1 Preferences and leisure

$$v^i(c_t, l_t, \eta_t) = \frac{((c_t/\eta_t)^\omega l_t^{1-\omega})^{1-\gamma} - 1}{1-\gamma}, \quad l_t^{i,j} = L^{i,j} - n_t^i - \Delta_t^{i,j} \mathbf{1}\{n_t^i > 0\}.$$

Interpretation. The fixed cost of work makes participation an explicit extensive-margin choice.

3.2 Wages and human capital

$$\log \varepsilon_{t+1}^i = \rho_\varepsilon^i \log \varepsilon_t^i + \nu_t^i, \quad w_t^i = e_t^i(\bar{y}_t^i) \varepsilon_t^i.$$

Interpretation. Work today raises human capital and future wages. Thus marriage-related wedges have dynamic effects.

3.3 Marriage, divorce, and matching

Singles marry with probabilities depending on age, gender, and wage shocks. Matching is assortative in wage shocks and draws a partner's wealth and accumulated earnings. Couples face divorce risk and split assets equally in divorce.

Interpretation. Single women's labour supply responds to marriage-related rules because they anticipate future marriage and future benefit/tax treatment.

3.4 Couple problem

$$W_t^c = \max_{c, a', n^1, n^2} \left\{ u(c, l^1, l^2) + \beta(1 - \zeta)EW_{t+1}^c + \beta\zeta \sum_i EW_{t+1}^s(a'/2) \right\}.$$

Interpretation. Both spouses choose labour supply jointly, so reforms can raise married women’s work and lower older married men’s work.

3.5 Policy counterfactuals

Policy effect = simulated outcome under reform – simulated outcome under benchmark.

Interpretation. The paper’s estimates are structural counterfactuals rather than reduced-form estimates from an observed reform.

4 Identification and policy-counterfactual strategy

- First-step input processes are identified from PSID and HRS data: wage processes, marriage, divorce, matching, children, survival, health, medical expenses, and taxes.
- Preference and time-cost parameters are identified by matching participation, hours, and wealth moments across age, gender, and marital status.
- Model validation uses wage correlations within couples, couple work-type shares, labour supply elasticities, and wealth profiles.
- Policy effects are generated by simulating the same households under benchmark and counterfactual rules.
- Robustness checks vary marriage/divorce rates and home-production time.

Interpretation. The identifying variation is the model’s ability to match a rich set of moments and then use the estimated primitives to run internally consistent counterfactuals.

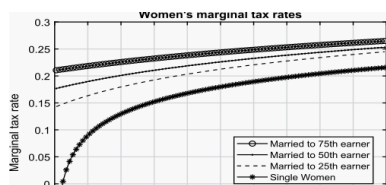
5 Tables and figures

5.1 Figure 1: Women’s marginal tax rates as a function of earnings + Figure 2: Average Social Security benefits

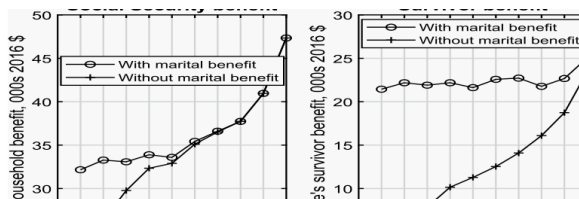
Read: Figure 1 shows that married women face higher marginal tax rates than single women. Figure 2 shows that spousal and survivor benefits are large for low-own-benefit women.

Interpretation. Joint taxation lowers the current return to work; Social Security marital benefits lower the retirement return to own work.

Economic message. These two wedges motivate the policy counterfactuals.



(a) Women’s marginal tax rates

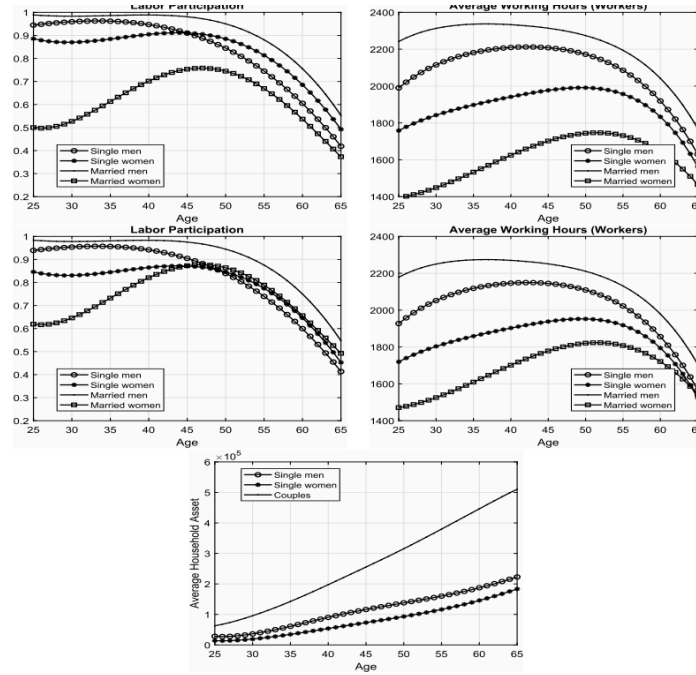


(b) Social Security benefits

5.2 Figure 3: Life-cycle profiles by gender and marital status

Read: The figure shows participation, hours, and wealth by gender-marital group for the 1945 and 1955 cohorts.

Interpretation. The 1955 cohort has higher married-women participation, but the model still finds large policy effects.



Life-cycle profiles

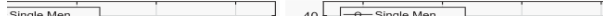
5.3 Table 1: First-step estimated inputs summary + Figure 4: Average wage profiles

Read: Table 1 lists the estimated input processes; Figure 4 shows wage profiles.

Interpretation. The wage profiles and first-step processes form the exogenous environment for the structural model.

TABLE 1
First-step estimated inputs summary

Estimated processes		Source
Wages		
$e_t^j(\cdot)$	Endogenous age-efficiency profiles	PSID
ϵ_t^j	Wage shocks	PSID
Demographics		
$s_t^{i,j}(\psi_t^j)$	Survival probability	HRS
$\zeta_t(\cdot)$	Divorce probability	PSID
$v_t(\cdot)$	Probability of getting married	PSID
$\xi_t(\cdot)$	Matching probability	PSID
$\theta_t(\cdot)$	Partner's wealth and earnings	PSID
$f^{0,5}(i,j,t)$	Number of children ages 0–5	PSID
$f^{6,11}(i,j,t)$	Number of children ages 6–11	PSID
Health shock		
$m_t^{i,j}(\psi_t^j)$	Medical expenses	HRS
$\pi_t^{i,j}(\psi_t^j)$	Transition matrix for health status	HRS
Government policy		
λ_t^j, τ_t^j	Income tax	See text



(a) First-step inputs

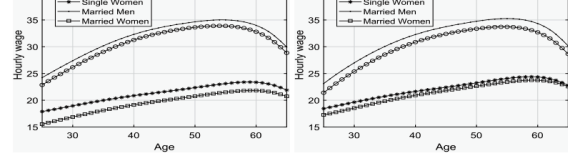


FIGURE 4
Average wage. Left: 1945 cohort. Right, 1955 cohort. PSID.

In the second step, we use the MSM. For the 1945 cohort, we normalize the time endowment

(b) Average wage profiles

5.4 Table 2: Second-step parameters + Figures 5 and 6: Model fit

Read: Table 2 gives estimated preferences and time costs. Figures 5 and 6 show that the model matches participation, hours, and wealth.

Interpretation. The joint fit of labour and savings is central for credible policy and welfare analysis.

TABLE 2 Second-step estimated model parameters		
Estimated parameters	1945 cohort	1955 cohort
β : Discount factor	0.990	As 1945 cohort
ω : Consumption weight	0.407	As 1945 cohort
$L^{2,1}$: Time endowment (weekly hours), single women	107	112
$L^{2,2}$: Time endowment (weekly hours), married men	107	100
$L^{2,3}$: Time endowment (weekly hours), married women	88	88
$\tau^{5,5}$: Prop. child care cost for children age 0-5	28%	22%
$\tau^{6,11}$: Prop. child care cost for children age 6-11	7%	19%
Φ_1^p : Partic. cost	Figure 10 in Supplementary Appendix G	Figure 10 in Supplementary Appendix G

(a) Second-step parameters

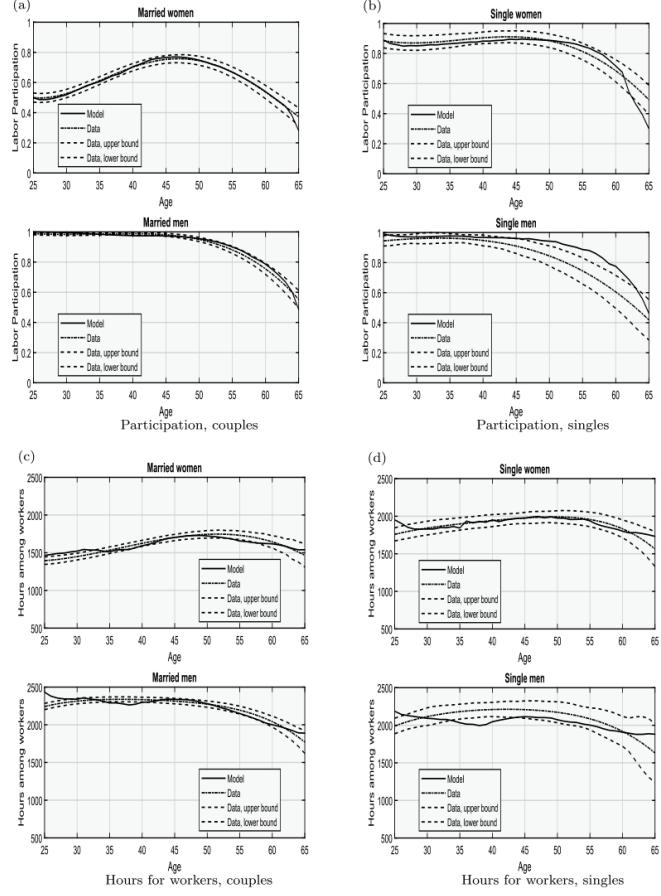
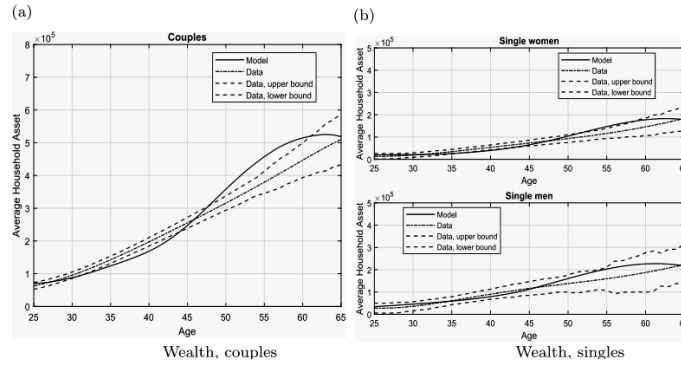


FIGURE 5

Model-implied participation and hours, and wages and CES confidence intervals from the BEPS

(b) Participation and hours fit



Wealth fit

5.5 Table 3: Correlation of husbands' and wives' wages + Figure 7 and Table 4: Validation and elasticities

Read: The model broadly matches wage correlations within couples and work-type shares. Elasticities are plausible relative to the literature.

Interpretation. These checks validate the within-couple labour-supply channel.

TABLE 3
Correlation of husbands and wives' wages at various ages, workers, model and data, 1945 cohort

Age group	Model	Data
25-34	0.415	0.338
35-44	0.376	0.314
45-54	0.385	0.300

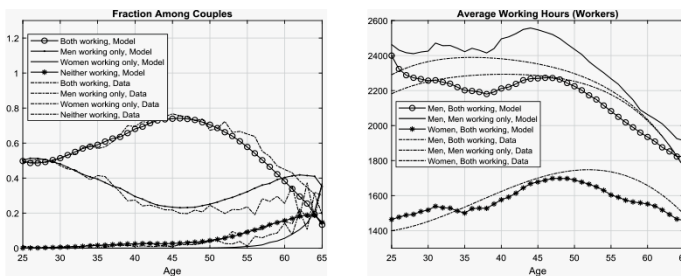


FIGURE 7
Participation (left) and working hours (right) for people in couples. Model (solid) and PSID (dashed). 1945 cohort.

(a) Wage correlations

(b) Couple validation

	Participation				Hours among workers			
	Married		Single		Married		Single	
	W	M	W	M	W	M	W	M
0	1.0	0.0	0.5	0.2	0.2	0.3	0.4	0.4
0	0.7	0.1	0.4	0.2	0.4	0.5	0.4	0.5
0	0.6	0.2	0.4	0.5	0.4	0.5	0.8	0.5
0	1.1	0.8	1.8	1.4	0.3	0.3	0.5	0.4

FIGURE 8: WAGE INCREASE AND PARTICIPATION ELASTICITIES

Labour supply elasticities

5.6 Figure 8: Wage increase and participation elasticities

Read: A 5% married-women wage increase raises their participation and lowers married men's participation after age 40.

Interpretation. The figure demonstrates within-couple substitution and strong female extensive-margin responsiveness.

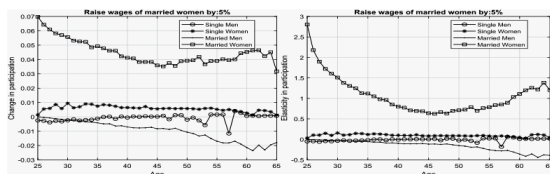


FIGURE 8
Change in participation (left) and implied participation elasticities (right) for a 5% permanent increase in the wage

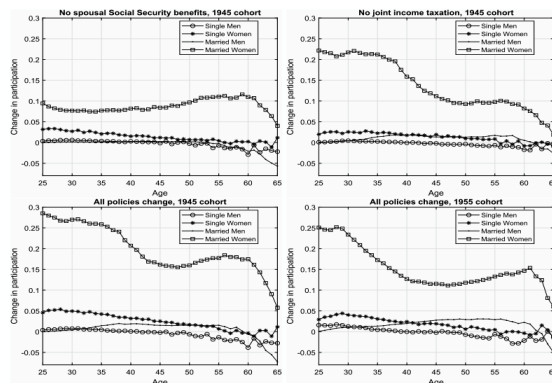


FIGURE 9
Changes in participation as a result of the elimination of various marriage-related policies under government budget balance.

5.7 Figure 9: Policy participation changes + Tables 5 and 6: Wage quintiles and wealth at age 66

Read: Removing spousal benefits, joint taxation, or all marital provisions raises married women's participation and retirement wealth.

Interpretation. Table 5 explains why effects are large: women are concentrated in lower wage quintiles and are often secondary earners.

TABLE 5
Distribution of men and women at age 25 across wage quintiles, 1945 (top) and 1955 cohort (bottom)

	Wage quintile (%)				
	1	2	3	4	5
1945 cohort					
Men	7.9	12.3	21.7	29.1	29.1
Women	32.5	27.9	18.3	10.7	10.7
1955 cohort					
Men	10.1	14.3	20.0	25.5	30.1
Women	28.5	24.9	20.2	15.1	11.4

Notes: PSID data.

(a) Wage quintiles

TABLE 6
Change in wealth at age 66, in percentages, as a result of each policy change when the income tax is reduced to balance the government budget

	Couples (%)	Single men (%\$)	Single women (%)
1945 cohort, no spousal Social Security benefits	14.6	8.5	11.2
1945 cohort, no marital-related policies	19.5	10.1	15.4
1955 cohort, no marital-related policies	17.9	10.0	13.7

(b) Wealth at age 66

5.8 Table 7: Welfare compensation

Read: Balanced-budget reforms produce large gains for couples and single men and smaller/heterogeneous effects for single women.

Interpretation. Welfare gains are largest when all marriage-related provisions are removed and the surplus lowers income taxation.

TABLE 7
Asset compensation required for staying in the benchmark economy, normalized as a fraction of average income

	All			Winners			Losers		
	Couples	SW	SM	Couples	SW	SM	Couples	SW	SM
1945 cohort									
(1) Remove Social Security spousal benefits, unbalanced budget									
Average	-0.24	-0.20	0.25	0.00	0.00	0.25	-0.24	-0.20	-0.02
Percentage				0.0	0.0	100.0	100.0	100.0	0.0
(2) Remove Social Security spousal benefits, balanced budget									
Average	0.66	0.19	1.15	0.66	0.20	1.15	0.00	-0.03	0.00
Percentage				100.0	92.5	100.0	0.0	7.5	0.0
(3) Remove joint income taxation, unbalanced budget									
Average	0.06	-0.18	0.81	0.29	0.06	0.81	-0.19	-0.19	0.00
Percentage				52.8	4.9	100.0	47.2	95.1	0.0
(4) Remove joint income taxation, balanced budget									
Average	0.31	-0.08	1.06	0.42	0.12	1.06	-0.09	-0.13	0.00
Percentage				78.8	20.6	100.0	21.2	79.4	0.0
(5) Remove all marital-related policies, balanced budget									
Average	0.80	0.05	1.97	0.80	0.33	1.97	-0.03	-0.12	0.00
Percentage				98.8	37.4	100.0	1.2	62.6	0.0
1955 cohort									
(6) Remove all marital-related policies, balanced budget									
Average	0.73	0.21	1.14	0.74	0.30	1.14	-0.04	-0.04	-0.03
Percentage				98.2	74.1	100.0	1.8	25.9	0.0

Notes: Top line for each experiment: average welfare gain or loss. Bottom line for each experiment: fraction in that group

Welfare compensation

5.9 Figure 10: Marriage/divorce robustness + Figure 11 and Table 8: Home-production robustness

Read: Results are robust to large marriage/divorce changes; allowing more home-production flexibility raises female participation and wealth effects.

Interpretation. The benchmark is conservative regarding home-production flexibility.

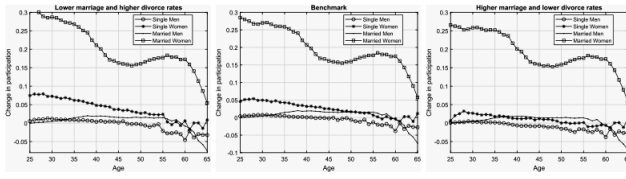


FIGURE 10
Changes in participation as a result of the elimination of all marriage-related policies when marriage and divorce patterns also change with policy. Government budget balance, 1945 cohort.

(a) Marriage/divorce robustness

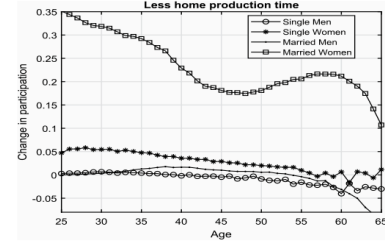


FIGURE 11
in participation as a result of the elimination of all marriage-related policies when available time increases by 3 h each week as policy changes. Government budget balance, 1945 cohort.

(b) Home-production robustness

TABLE 8
Change in wealth at age 66 after removing all marriage-related policies when available time increases by 3 h each week as policy changes

	Couples (%)	Single men (%)	Single women (%)
Benchmark	19.5	10.1	15.4
Reduce home production	21.5	10.8	16.8

Notes: Government budget balance, 1945 cohort.

Home-production wealth robustness

6 Interpretive synthesis

The paper is a dynamic wedge-accounting exercise. Joint taxation lowers the current return to secondary-earner labour. Spousal and survivor benefits lower the retirement return to own work. Human capital makes these distortions persistent. Savings and welfare respond because households face retirement, health, and survival risk.

7 Short summary

- Marriage-related provisions strongly discourage female labour supply.
- Removing them raises married women's participation, increases retirement wealth, and generates welfare gains for most households.
- Effects remain large for the 1955 cohort, despite higher female participation.
- Robustness checks do not overturn the results.

8 Conclusion

8.1 Core conclusion

Marriage-related taxes and Social Security benefits materially depress female labour supply and reduce savings.

8.2 Economic intuition

The policies insure spouses but create implicit taxes on secondary earners and weaken incentives for own human-capital accumulation.

8.3 Takeaway

Individualizing tax and retirement-benefit rules can raise female labour supply and welfare, while distributional effects on low-human-capital women require attention.